

OPTIMAL PORTFOLIO ANALYSIS IN THE ENERGY SECTOR STOCKS USING THE MARKOWITZ MODEL

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ABSTRACT

Introduction: Currently, the energy sector is attracting a lot of investors, as european countries are facing an energy crisis due to the impact of the Russia-Ukraine war since the beginning of 2022. This event has had a significant influence on the stock index of the energy sector. Some stocks in the energy sector in the Indonesian capital market that have experienced price increases are PT AKR Corporindo Tbk. (AKRA), PT Adaro Minerals Indonesia Tbk (ADMR), and PT Medco Energi Internasional Tbk. (MEDC).

Methods: This study aims to obtain the optimum allocation for three stocks in the energy sector. The method used to obtain the optimum allocation is the Markowitz Model with stock price data from the daily historical data from March 27, 2023, to March 22, 2024.

Results: The research results show that AKRA stock has the highest proportion at 56.1%, while ADMR is at 23.2%, and MEDC is at 20.6%, with an expected portfolio return of 0.0008 and a risk of 0.04282.

Conclusion and suggestion: AKRA stock is predicted to generate the lowest profit compared to ADMR and MEDC. MEDC stock has the highest risk; it has the highest standard deviation value, indicating that this stock has the highest return fluctuation risk compared to AKRA and ADMR. The resulting portfolio has relatively stable profit potential with a moderate level of risk. By utilizing the Markowitz model, optimal investment weights for each stock have been generated, thus investors can consider their portfolio allocation based on these analysis results, taking into account the balance between expected return and risk.

INTRODUCTION

In the current global economic environment, investment in the energy sector is a primary focus for investors. The energy sector is the backbone of the global economy, playing a vital role in supporting the growth and development of various industries. Energy stocks are shares of companies that trade products and services related to both non-renewable and renewable energy extraction. Their revenue is directly influenced by global commodity prices, such as crude oil, natural gas, and coal mining. From a fundamental analysis perspective, these companies represent leading commodities whose products are needed by both domestic and international markets (Sipayung et al., 2023).

There are various factors driving significant growth in the energy sector stock index. One of them is geopolitical tension between Russia and Ukraine, which has led to spikes in energy commodity prices. According to Supriyanto (2012), Russia and Ukraine are countries that play important roles in global energy and food production. Russia, in particular, is known as one of the three largest oil producers in the world, after the United States and Saudi Arabia (Huka & Kelen, 2022). However, oil embargo sanctions imposed by the European Union and Group of Seven (G7) countries against Russia have caused a shortage in global oil supply and significantly increased oil prices.

In this context, obtaining an optimal portfolio in the energy sector is important to consider for investors. Several companies in the energy sector operating in Indonesia are PT AKR Corporindo Tbk. (AKRA), PT Adaro Minerals Indonesia Tbk (ADMR), and PT Medco Energi Internasional Tbk. (MEDC). These three companies are major players in the Indonesian energy sector, each involved in energy distribution and trading, oil and gas exploration and production, as well as coal mining.

A portfolio is a form of investment in various companies formed from a combination of assets owned by these companies to achieve optimal results with minimal risk (Kusumawati & Subekti, 2013). Determining an efficient portfolio is a key aspect to consider in seeking an optimal portfolio. Before determining the optimal portfolio, the first step is to form an efficient portfolio (Suprihatin and Budiyanto, 2014).

The Markowitz Model is one of the appropriate models in portfolio selection, emphasizing efforts to maximize expected returns and reduce stock uncertainty or risk. This study aims to conduct an analysis of optimal portfolios using the Markowitz Model

with the Lagrange Method. The Lagrange Method is a method used to determine the maximum or minimum value of a function subject to certain constrained conditions.

This research is relevant to the study previously conducted by Plessis and Ward (2009) at the Johannesburg Securities Exchange (JSE). The Markowitz Model can be used as a portfolio formation model because it is capable of reducing risk and increasing returns over the long term. Another study was also conducted by Azizah et al., (2017) using the Markowitz Model and Single Index Model (a study on stocks listed on the Jakarta Islamic Index (JII)), which concluded that the optimal combination of investments in stocks listed on the Jakarta Islamic Index based on the Markowitz model yielded eight stocks.

LITERATURE REVIEW

Capital Market

The capital market is a market for various long-term financial instruments that can be traded, including debt securities (bonds), equity (stocks), mutual funds, derivative instruments, and other instruments (Nurdianingsih & Suryadi, 2021). In the capital market, investing in stocks is one of the primary choices for investors seeking optimal returns. The capital market plays a crucial role in the economy as companies can access the funds needed to grow and expand. Meanwhile, investors have the opportunity to invest in a diverse range of financial instruments to achieve their financial goals (Septiani, D., et al 2020).

Investment

Investment is a commitment of funds made at present to gain profits in the future (Tandelilin, 2010: 2). Investment is the act of deploying capital with the expectation of earning returns in the future. Through investment in the capital market, individuals allocate capital or assets into investment instruments available in the market, either to meet long-term or short-term needs. Investing in stocks involves purchasing ownership stakes in a company. Investors who buy stocks become shareholders and have claims to the company's income and assets. Profits from stock investments can be obtained through the appreciation of stock prices (capital gain) and dividend distributions. Investing in stocks provides opportunities to grow capital with risks commensurate with the performance of the respective companies (Purnamawati, I.G.A., et al 2023).

Optimal Portfolio

A portfolio is a form of investment in various companies formed from a combination of assets owned by these companies to achieve optimal results with minimal risk. Determining an efficient portfolio is a key aspect to consider in seeking an optimal

portfolio. Before determining the optimal portfolio, the first step is to form an efficient portfolio (Subekti, R et al., 2022).

Markowitz Model

The Markowitz Model is one of the appropriate models in portfolio selection, emphasizing efforts to maximize expected returns and reduce stock uncertainty or risk. This study aims to conduct an analysis of optimal portfolios using the Markowitz Model with the Lagrange Method. The Lagrange Method is a method used to determine the maximum or minimum value of a function subject to certain constrained conditions (Subekti, R et al., 2022).

RESEARCH METHODS

The data used in this study are secondary data accessible on the Yahoo Finance website (www.finance.yahoo.com) encompassing daily closing prices of three stocks: PT AKR Corporindo Tbk. (AKRA), PT Adaro Minerals Indonesia Tbk (ADMR), and PT Medco Energi Internasional Tbk. (MEDC). The research period spans from 26 April 2023 to 23 April 2024, resulting in 12 monthly return data entries for each company. The software utilized for data analysis in this study is RStudio, with the tseries package aiding in the time series analysis of stock prices and the PerformanceAnalytics package employed to identify risks and measure portfolio performance.

Return

Return is the result of the time factor and the risk involved in an investment, divided into two main components: Yield and Capital Gain. Yield refers to the periodic income generated from an investment, whereas Capital Gain is the profit obtained from the increase in the asset's value over time. These two components combined form the total return of the investment. This equation provides a framework for assessing the level of return obtained from a stock investment in more detail as follows (Hartono & Jogiyanto, 2014):

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}} \quad (1)$$

where,

R_t = Stock return rate in period t

P_t = Closing price of the stock in period t

P_{t-1} = Closing price of the stock in period $t - 1$

Expected Return & Risiko Portofolio

Expected return is the predicted outcome that will occur in the future, based on previously obtained results in percentage form. To calculate the expected return rate, the following equation can be used (Hartono & Jogiyanto, 2014):

$$E(R_i) = \frac{\sum_{i=1}^n R_{it}}{n} \quad (2)$$

where,

$E(R_i)$ = Expected return of the stock

R_i = Return of the stock in period

n = Number of observation periods

Risk can be defined as the possibility of a difference between the actual return and the expected return. The statistical measure of standard deviation is typically used to measure risk; the larger the standard deviation of a return distribution, the higher the investment risk (Husnan, 2012:192). Stock risk can be calculated using the following equation (Hartono & Jogiyanto, 2014):

$$\sigma_i = \sqrt{\frac{\sum_{i=1}^n (R_{it} - E(R_i))^2}{n - 1}} \quad (3)$$

where,

σ_i = Stock standard deviation

$E(R_i)$ = Expected return of the stock

R_{it} = Stock return rate in period t

n = Number of observation periods

Model Mean-Variance Markowitz

According to Harry Markowitz, the risk in a portfolio is influenced by the weighted average of the risks of individual assets and the covariance between the assets that make up the portfolio. Markowitz illustrated the principle of "don't put your eggs in one basket," suggesting that placing all investments in a single asset can significantly increase risk. In this context, if one asset experiences a large loss, the entire portfolio can be affected. To determine the optimal portfolio, Markowitz used the concepts of expected return and risk, where risk is measured by the variance of returns. This approach aims to minimize portfolio variance for a given level of expected return. Thus, the Markowitz mean-variance model provides a systematic framework for managing risk and designing a balanced investment portfolio. The Markowitz mean-variance model can be represented as follows:

$$\underset{w}{\text{minimize}} \quad \sigma_p^2 = \sum_{i=1}^n \sum_{j=1}^n w_i w_j \sigma_{ij} \quad (4)$$

where:

σ_p^2 = Portfolio variance

w_i = Weight proportion of stock i

w_j = Weight proportion of stock j

σ_{ij} = Covariance between stocks

$E(R_i)$ = Expected return of stock

R_p = Portfolio return

Optimasi Portofolio

To optimize the optimal portfolio, it is necessary to determine the formulation of the optimal weight proportions and portfolio risk by optimizing the Markowitz mean-variance model using the Lagrange method. However, before that, we will transform this formulation into matrix form to simplify the calculations (Subekti, R et al., 2022).

$$w = \lambda_1 \Sigma^{-1} \mu + \lambda_2 \Sigma^{-1} \mu \quad (5)$$

To address this constrained problem, it can be solved by forming the Lagrange function, resulting in:

$$L = \frac{1}{2} \mathbf{w}' \Sigma \mathbf{w} - \lambda_1 (\mathbf{w}' \boldsymbol{\mu} - \mu_p) - \lambda_2 (\mathbf{1}' \mathbf{w} - 1) \quad (6)$$

RESULT AND ANALYSIS

The daily stock data of the three stocks can be seen in Figure 1. The curve shows that the stocks of PT AKR Corporindo Tbk. (AKRA), PT Adaro Minerals Indonesia Tbk (ADMR), and PT Medco Energi Internasional Tbk. (MEDC) experienced a fairly significant price increase.

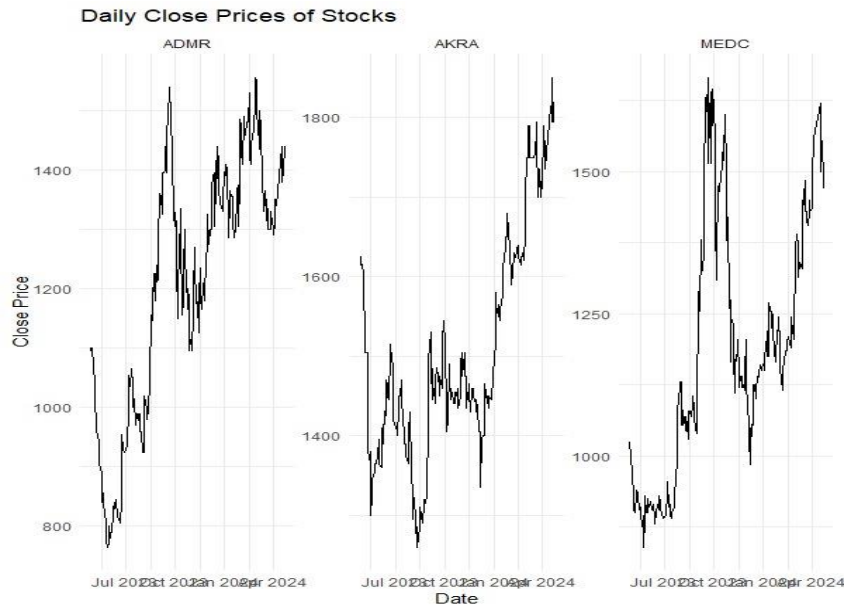


Figure 1. Daily Close Price Data for Each Stock

Source: Stock Data Processing Results (2023-2024)

After obtaining the daily data of these three stocks, the next step is to generate return data. The data used in this study is time series return data. Return of an asset is the rate of return or the result obtained from an investment. Return can reflect actual price changes. The curve of daily stock return data can be seen in Figure 2.

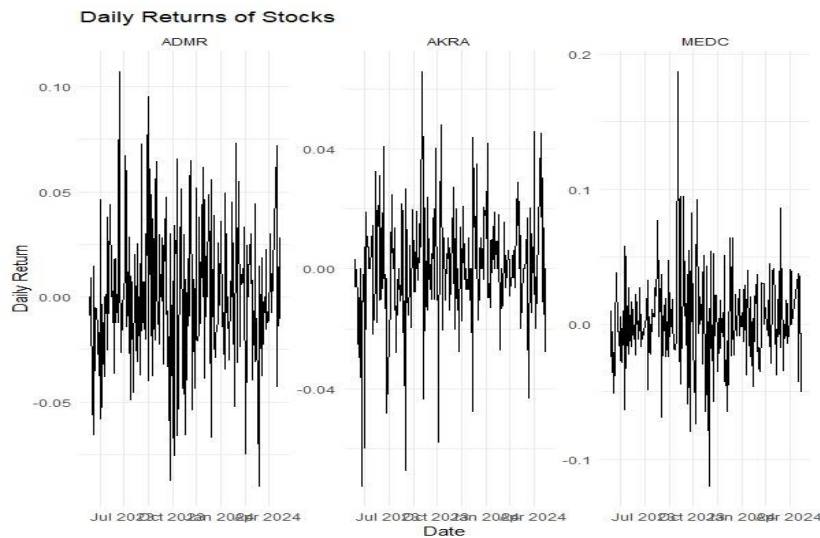


Figure 2. Daily Return Data for Each Stock

Source: Stock Data Processing Results (2023-2024)

In Figure 2, it shows the calculation of daily closing stock returns from three companies. The purpose of calculating stock returns is to depict the profits gained.

After obtaining the returns, the next step is to calculate the expected return, variance, and standard deviation. Expected return explains the level of expected profit that may be obtained from each stock, which can be positive or negative. Standard deviation and variance are used to calculate stock risk, measuring the absolute deviation of values that have occurred from their expected values. The results of the expected return, standard deviation, and variance calculations for each company's stock are presented in Table 1.

Table 1. Expected Return, Variance, Standard Deviation of Each Stock

Stock	<i>Expected Return</i>	Variance	Standard Deviation
AKRA	0,0004252017	0,0003404731	0,01845191
ADMR	0,0011062463	0,0011304640	0,03362237
MEDC	0,0015408965	0,0012238145	0,03498306

Source: Stock Data Processing Results (2023-2024)

In Table 1, it shows three stocks of companies, where MEDC stock provides the highest expected return at 0.0015408965. Next, ADMR stock follows at 0.0011062463,

and AKRA stock provides the lowest expected return at 0.0004252017. All three stocks generate positive expected returns. This means that all three stocks can be included in determining the optimal portfolio. The stock with the highest risk is MEDC, seen from the standard deviation value of 3.4%, in line with the positive expected return obtained. Meanwhile, AKRA stock has the lowest risk with a standard deviation value of 1.8%. Similarly, in terms of variance, MEDC has the highest value, while AKRA has the lowest.

The next step is to calculate the variance-covariance matrix between stocks. The variance-covariance matrix between stocks indicates to what extent the returns of two stocks tend to move together.

Table 2. Variance-Covariance Matrix of Each Stock

Stock	AKRA	ADMR	MEDC
AKRA	0,0003404731	0,0002079146	0,0004441593
ADMR	0,0002079146	0,0022609279	0,0004368177
MEDC	0,0004441593	0,0004368177	0,0024476290

Source: Stock Data Processing Results (2023-2024)

In Table 2, it can be seen that all covariances are positive. Positive covariance values mean that if the return of one security increases, the other security also tends to increase. Negative numbers indicate that if the return of one security moves up, the other security will move down, and vice versa for zero values, which means the movement of one security is independent of the other security. Covariance values can be used by investors to maximize the expected return of the desired portfolio.

To obtain an optimal portfolio, it is necessary to calculate the expected return, standard deviation, variance, and investment weights. The results of the optimal weight calculation using equation (5) are presented in Table 3. The optimal portfolio allocation is 56.1% for AKRA stock, 23.3% for ADMR stock, and 20.6% for MEDC stock.

Table 3. Stock Portfolio Optimization Weights

	AKRA	ADMR	MEDC
Optimal Investment Weights	0,5610815	0,2326915	0,2062261

Source: Stock Data Processing Results (2023-2024)

The results of the Markowitz portfolio optimization calculation are presented in Table 4. The portfolio's expected return generated is 0.0008137599. The obtained variance is 0.001377873. Meanwhile, the standard deviation is 0.04282606. A lower standard deviation indicates lower volatility in the portfolio. The results indicate a well-diversified portfolio with moderate expected return and relatively low risk.

Table 4. Markowitz Portfolio Optimization Calculation

	Model Markowitz
<i>Expected Return</i>	0,0008137599
Variance	0,001377873
Standar Deviation	0,04282606

Source: Stock Data Processing Results (2023-2024)

CONCLUSION

Based on the portfolio analysis of stocks using the Markowitz model for PT AKR Corporindo Tbk. (AKRA), PT Medco Energi Internasional Tbk. (MEDC), and PT Adaro Minerals Indonesia Tbk (ADMR) for the period from April 26, 2023, to April 24, 2024, it was found that AKRA offers the lowest expected return. Among the three analyzed stocks, AKRA has the lowest expected return at 0.0004252017. This indicates that statistically, AKRA is predicted to generate the lowest profit compared to ADMR and MEDC. MEDC stock has the highest risk, with the highest standard deviation value of 0.03498306, indicating that this stock has the highest fluctuation risk in returns compared to AKRA and ADMR.

Portfolio optimization resulted in optimal investment weights for each stock: AKRA yielded a weight of 56.1%, ADMR at 23.2%, and MEDC at 20.6%. This optimal portfolio is expected to yield an expected return of 0.0008137599 with a variance of 0.001378 and a standard deviation of 0.04282606. This means that the portfolio has the potential for relatively stable profits with a moderate level of risk.

By utilizing the Markowitz model, optimal investment weights for each stock have been generated, thus investors can consider their portfolio allocation based on these analysis results, taking into account the balance between expected return and risk. For further research, other methods related to risk modeling can be employed.

REFERENCES

- Adaro Minerals Indonesia. (2024). Pada Yahoo Finance. Diambil 23 April 2024, dari <https://finance.yahoo.com/quote/ADMR.JK>
- AKR Corporindo. (2024). Pada Yahoo Finance. Diambil 23 April 2024, dari <https://finance.yahoo.com/quote/AKRA.JK>
- D. Ruppert. (2010) *Statistics and Data Analysis for Financial Engineering* (New York: Springer)
- Elton, E. J., and Gruber, M. J. (2014). *Modern Portfolio Theory and Investment Analysis*, 9th Edition, John Wiley & Sons, New York
- Francis, J. (1991). *"Investment: Analysis and management"*. 5th edition. Singapore: Mc.Graw Hill Inc
- Ketut Sinta Trisnadewi and Nyoman Angga Pradipa. (2022). 'Analisis Studi Peristiwa Invasi Rusia ke Ukraina terhadap Harga Saham IDX 80 di Bursa Efek Indonesia', *Jurnal Ekonomi Bisnis dan Akuntansi*, 3(1), pp. 214–223. doi:10.55606/jebaku.v3i1.1427.
- Kusumawati, N., & Subekti, R. (2013). Aplikasi Pembentukan Portofolio Saham Lq-45 Estimasi Theil Mixed. *Prosiding Seminar Nasional Matematika Dan Pendidikan Matematika, November*, 192–198.
- Markowitz, H. (1952). Portofolio Selection. *The Journal of Finance*, Vol.7, No.1. McGraw Hill, New York. [11] Wei, W.W.S. 1990. *Time Series Analysis: Univariate and Multivariate Methods*. AddisonWesley Publishing Company, USA.
- Medco Energi Internasional. (2024). Pada Yahoo Finance. Diambil 23 April 2024, dari <https://finance.yahoo.com/quote/MEDC.JK>
- Moerdianto, R. M., Anto, A. S. U., Gazali, Y. P., Kartini, K. K., & Rura, Y. R. (2022). Pengaruh Return on Asset Dan Earning Per Share Terhadap Harga Saham Perusahaan Sektor Energi Yang Terdaftar Di Bursa Efek Indonesia. *Eqien - Jurnal Ekonomi Dan Bisnis*, 11(04). <https://doi.org/10.34308/eqien.v11i04.1237>
- Nurdianingsih, R., & Suryadi, E. (2021). Analisis Perbandingan Portofolio Optimal Dengan Menggunakan Model Single Indeks Dan Model Markowitz Dalam Penetapan Investasi Saham. *Jurnal Produktivitas*, 8, 46–55. www.openjurnal.unmuhpnk.ac.id/index.php/jp
- Peterson BG, Carl P (2020). *_PerformanceAnalytics: Econometric Tools for Performance and Risk Analysis_*. R package version 2.0.4, <<https://CRAN.R-project.org/package=PerformanceAnalytics>>.
- Plessis, AJ Du & Ward. (2009). A Note on Applying the Markowitz Portfolio Selection Model as a Passive Investment Strategy On The JSE. *Investment Analysts Journal*, 9 (69), 39-46.
- Purnamawati, I.G.A. et al. (2023) 'Family Ownership Matters for Investors in Indonesia's Mining Companies', *International Journal of Financial and Investment Studies (IJFIS)*, 4(1), pp. 17–30. doi:10.9744/ijfis.4.1.17-30.

- Septiani, D. and Lilis Karlina, A.M.F. (2020) 'Pengenalan Manajemen Investasi Dan Pasar Modal Bagi Siswa/I Dan Guru Akuntansi Smk Bintang Nusantara', KUAT : Keuangan Umum dan Akuntansi Terapan, 2(1), pp. 58–63. doi:10.31092/kuat.v2i1.681.
- Sipayung, F. H., Puspitawati, E., Rembulan Nurdianto, N., & Ismail Khozin, M. (2023). Analisis Faktor-Faktor yang Mempengaruhi Indeks Saham Sektor Energi di Indonesia. *Desember*, 2023(1), 62–81. <https://elegis.universitaspertamina.ac.id/index.php/home1/article/view/4>
- Subekti, R., Abdurakhman and Rosadi, D. (2022) 'The efficiency frontier of markowitz and black-litterman model: A case study on the shariah-compliant stock in Jakarta Islamic Index', *AIP Conference Proceedings*, 2575(December). doi:10.1063/5.0108225.
- Subekti, R., Abdurakhman and Rosadi, D. (2021) 'Reverse optimization and capital asset pricing model in the application of the Black Litterman portfolio', *Journal of Physics: Conference Series*, 1918(4), pp. 0–6. doi:10.1088/1742-6596/1918/4/042037.
- Subekti, R., Abdurakhman and Rosadi, D. (2022) 'Toward the Black–Litterman with Shariah-compliant asset pricing model: a case study on the Indonesian stock market during the COVID-19 pandemic', *International Journal of Islamic and Middle Eastern Finance and Management*, 15(6), pp. 1150–1164. doi:10.1108/IMEFM-12-2020-0633.
- Supandi, E. D. (2016). Karakteristik Kurva Efisien Frontier dalam Menentukan Portofolio Optimal. *Jurnal Teknik Industri*, 18(1). <https://doi.org/10.9744/jti.18.1.43-50>
- Supriyanto and Mulyantini, S. (2012) 'Faktor Yang Berpengaruh Indeks Harga Saham Pada Bursa Efek Indonesia Tahun 2009 - 2012', *Ekonomi dan Bisnis*, 3(2), pp. 115–132. doi:10.35590/jeb.v3i2.730.
- Tandelilin, E. (2010). Portofolio dan Investasi: Teori dan Aplikasi. 07. Yogyakarta: Kanisius.
- Trapletti A, Hornik K (2023). *_tseries: Time Series Analysis and Computational Finance_*. R package version 0.10-54, <<https://CRAN.R-project.org/package=tseries>>.
- Wrede, R. C. dan M. Spiegel. (2002). *Schaum's Outlines: Advanced Calculus*. 2nd Edition.